OAL — Network Defense Intelligence Exercise

Collaborative Research Review — Cycle 1

Independent critical assessment by Perplexity

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Target: OAL-NDIE-RESEARCH-EXEC-CHATGPT-001, v1.0, dated November 5, 2026

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Prepared for: Nathan Lewis and the six-system collaborative AI research team

Deliverables: matching PDF and DOCX; no target edits, testing, deployment or GitHub writes performed.

Review scope, independence and method

This review assesses the supplied ChatGPT-authored execution report, not Perplexity's earlier 25-page report. The inventory also includes other systems' research reports; they were not read for this Cycle 1 assessment. Prior peer prompts and Perplexity's own earlier work are known, so this is a source-rechecking independent judgment, not a blind review with no prior exposure. No other system's review is represented as completed or endorsed.

Methods: direct reading of the ten-page target; selected official/publication-page retrievals; targeted searches for a difficult-to-access cited review and missing empirical evidence; inspection of the related Project File disclaimer; and independent Python arithmetic checks. No laboratory experiment, benchmark, human study, funder contact, vendor quote or full systematic literature review was performed. Access date for rechecked online sources: November 5, 2026. Review source identifiers R01-R13 are separate from the target's S01-S18.

Verification categories used below: verified publication and cited proposition; partly verified (publication/abstract but not full paper/count); not rechecked in this cycle; or unavailable to this reviewer. Not rechecked does not mean false. Framework existence is not efficacy evidence, and source authenticity is not independent replication.

A. Executive assessment

Overall recommendation: REVISE AND NARROW, then conditionally proceed with a bounded measurement study. Do not approve the target as a validated architecture, a confirmatory experiment protocol or an investment-ready budget. Confidence: high that the scope and methodological corrections are necessary; moderate that a carefully scoped study could be worthwhile; low in comparative effectiveness and specific future operating costs.

The target's main conclusion is credible: the national, seamless, every-attacker version is not established by the cited evidence, and genuine-asset outcomes matter more than decoy contacts. Its separation of creative canon, hypotheses and demonstrated tools is an important strength. Cowrie, NIST's resource-centric model and MITRE's decoy vocabulary substantiate components, not the entire chain. [Target §§1-6; R01-R03]

Four material findings require attention before another iteration:

1. Critical experimental identifiability issue: the report simultaneously requires an independent invariant that the test attacker can never obtain an unauthorized route to the protected synthetic asset and proposes unauthorized reachability as a treatment outcome. If the invariant fixes that outcome at zero in every arm, E1 cannot estimate protection benefit. Separate external research safety from deliberately challengeable synthetic mission security. [Target §§4,9; reviewer inference]

2. Definite budget arithmetic error: the stated 75%-150% sensitivity rule is not applied to the published base figures. For the $588,000 proof of concept, that rule yields $441,000-$882,000, not approximately $397,000-$794,000. The same inconsistency affects the range and multi-organization stages. This is a reproducible calculation error, not a disagreement about price assumptions. [Target §10; Appendix 1]

3. The experimental null combines several endpoints without specifying which is primary or how multiple testing, censoring and nondetection are handled. Faster isolation among only detected attacks can reward a system that misses the difficult attacks. [Target §§2,9; reviewer inference]

4. Important citations exist and several numerical claims check out, but not all evidentiary detail was verified. The 2026 IEEE review exists at the stated identifier; the publisher page could not be retrieved, and the 97-study/performance-gap details are only corroborated through a secondary abstract mirror. Mark that limitation rather than certify full verification. [Target S15; R09]

No result here establishes a need for nationwide deployment or approves a new canonical interpretation. A small decision-quality study is the maximum justified next commitment.

B. Supported findings

B1. Established building blocks. Cowrie's current repository documents an emulated Unix shell, SSH/Telnet proxy observation, replayable sessions, JSON logs and an experimental LLM backend. It establishes that narrow protocol-specific deception and instrumentation exist. It does not establish transparent migration from arbitrary real authenticated services or undetectability. Freeze a release/commit in any experiment; a moving README is not a reproducible software version. [Target §3, S05; R03]

B2. Resource- and identity-centered protection. NIST SP 800-207 explicitly rejects implicit trust based solely on network location or ownership and describes authentication and authorization before resource sessions. This supports the baseline architecture and valid-access threat framing, not an assertion that buying a product guarantees zero trust. [Target §2, S04; R01]

B3. Deception vocabulary versus efficacy. MITRE D3FEND defines decoy resources and explicitly warns about resource exhaustion. Its mappings are described as inferred/experimental, and references include patents with a disclaimer that they do not establish product functionality. The target correctly calls the taxonomy nonexperimental; preserve that distinction throughout the capability matrix. [Target §§2,5; R02]

B4. Cyber Storm IX facts and governance relevance. The official AAR verifies April 2024, over 2,200 participants, cloud configuration/identity issues, and organizational coordination findings. Its evidence is exercise observations, questionnaires and debriefs, not a randomized deception evaluation. Its specific

lessons on provider responsibility, restoration SLAs, missing contacts and unavailable primary communications deserve stronger use in NDIE's governance tests. [Target §5, S07; R04, AAR pp.1,3-10]

B5. AIxCC count verified. DARPA's results page reports 54 synthetic vulnerabilities found and 43 patched across 63 challenges, as stated in the target. It additionally reports 18 non-synthetic vulnerabilities and 11 patches. The page's corrected denominator is important: prize competition performance is not a national deception trial, and challenge costs are not project lifecycle cost. [Target §5, S10; R05]

B6. Fingerprinting review exists. Chaudhry, Andersen and Dragoni's paper was published April 1, 2026, DOI 10.3390/fi18040190. The retrieved methods narrow the systematic inclusion to low-/medium-interaction systems and report 64 analyzed papers after snowballing. This supports detectability concerns, but not a universal ranking of every high-interaction or modern generative design. [Target S06; R06]

B7. LLM case is correctly cautious. Oliveira, Gomes and Almeida's August 31, 2026 paper reports 52 observations, a single evaluation run, 24 malicious and 28 benign observations, and preliminary proof-of-concept results. The target properly labels its external-validity limits. Subsequent use should also disclose the strong deterministic preprocessing/override layer and that an observation is a behavioral stage, not necessarily an independent campaign or person. [Target S16; R07]

B8. Funding ceilings partly verified. The NSF SaTC page confirms RES up to $1.2 million total/four years and SEED up to $300,000/two years. It states SEED is restricted to special topics defined by accompanying Dear Colleague Letters; it is not an unrestricted smaller RES grant. January 25 and September 27, 2027 are target dates, with proposals accepted anytime. [Target §11; R08]

B9. No optimum three seconds. The target is right not to claim an optimum or assume that key, route, identity, persona and visualization changes have the same effect. This review did not find direct validation of the exact national-scale cadence. That is a bounded search finding, not proof no classified or inaccessible work exists.

C. Disputed findings and overstatements

C1. Budget bands must be corrected. The four base calculations are internally consistent with their FTE, months, loaded-rate, direct-cost and 20% reserve formula. The narrative sensitivity examples are not. Correct values appear in Appendix 1. Keep unverified wages, loaded multipliers and reserves labeled; fixing arithmetic does not validate rates or economic benefit.

C2. "Auth, state, transport and app semantics prevent universal migration" is too categorical as technical explanation. The correct finding is that the report does not demonstrate arbitrary invisible migration under unbounded protocols and trust arrangements. For a precontrolled service with legitimate key/state handling, limited proxy continuity or explicit handoff can be feasible; the constraints are a supported scope, protocol and trust model. Do not turn lack of a general solution into proof that every bounded transfer is impossible. [Target §3; R03 supports proxy operation, not arbitrary production takeover.]

C3. Fail-closed mission isolation is not the same as safe research containment. A physical/logical outer lab boundary should never be breached. Inside it, synthetic mission assets must be challengeable if compromise is to be compared. Alternatively, retain perfect internal protection and explicitly study intelligence yield/workload, not reduction in already-zero reachability. Resolve this before implementing E1. [Target §§4,9]

C4. H0 conflates distinct reasons to continue. The report allows improved reachability, time-to-containment, useful intelligence or workload to justify the concept. That flexibility risks selecting a favorable outcome after inspection. A project could improve summaries but not protection; that can be useful, but requires a separate, narrower objective and business case. State one primary endpoint per study and distinguish protection, intelligence and analyst-assistance claims. [Target §2]

C5. The 1-percentage-point benign-diversion allowance is not an adequate default. On a workload of 10,000 benign sessions, one additional point could mean 100 extra diverted sessions. Whether acceptable depends on task criticality and reversible harm, not statistical significance alone. Bound both absolute rate and severity-weighted workflow harm; emergency workflows may require zero tolerated harmful diversion in the exercised cases. The target labels the number proposed, but does not justify it. [Target §9]

C6. A statistical nonsignificance result is not enough for a rational no-go. Section 13's language about benefit not distinguishable from baseline should distinguish an adequately precise negative result from an underpowered/inconclusive study. Use confidence intervals against a preregistered minimum worthwhile effect, or equivalence/noninferiority designs when those are the actual question. Neither "p > .05" nor no observed breach establishes equality.

C7. The 90-day plan conflates implementation, human-subject research and independent replication. It may be realistic for a small existing lab but is not justified without staffing, approvals and reusable infrastructure. Clock time for ethics/legal/partner access cannot be inferred from cloud provisioning time. Make dates conditional and allow E5 human/adaptive work to follow a separate gate. [Target §14]

C8. Current funding opportunity versus program presence remains insufficiently separated. SaTC caps verify; SEED requires topic fit. DARPA's BAA index and DOE/NIST SBIR program pages are discovery leads, not proof of a matching active award. Preserve uncertainty and add actual solicitation ID, deadline, eligibility and restrictions before any grant recommendation. This review does not certify DOE, NIST or DARPA availability. [Target §11; R08]

C9. The original threat evolution is too linear. Encounters 0-5 describe possible postures, not inevitable learning stages or a monotonic loss of deception value. A controlled study of over 130 professional red teamers found that the combination of decoys and disclosure had the greatest effect on measured forward progress. This does not prove deterrence, but contradicts a simple secrecy-is-required model. Branch the scenario into avoidance, disclosure effects, displacement, continued engagement and counter-deception. [Target §§6-7; R10]

D. Missing research

D1. Add primary causal evidence rather than only more reviews. Ferguson-Walter et al. (2021) directly manipulate decoy presence/disclosure. It is older than the report's 2026 reviews but more relevant to a human causal hypothesis. Add the study, its endpoints, randomization details and external-validity limits; do not copy its sample size as NDIE's power requirement. [R10]

D2. Add a concrete MTD physical-testbed counterweight. Kim et al., "Evaluating Performance and Security of a Hybrid Moving Target Defense in SDN Environments," DOI 10.1109/QRS57517.2022.00037, reports a physical SDN testbed, time-plus-IDS triggering and performance/security evaluation. An institutional bibliographic record verifies its abstract; full results were not independently reanalyzed here. This could qualify an overly broad "understudied" statement without establishing national scale. [R11]

D3. Newer verified survey lead: Casola, De Benedictis and Iorio's 2026 Edge-Cloud MTD systematic review is relevant to overhead and orchestration, but is neither the cited IEEE review nor interchangeable with it. Search extraction was available; full review/reanalysis remains backlog. Do not silently substitute it to repair an inaccessible citation. [R12]

D4. Quantify what the defender can actually see: TLS termination assumptions, service keys, client trust anchors, delegated identity, cached tokens, application transactions, external data copies and multi-tenant cloud management. A scope/eligibility table should show which sessions can even be considered for redirection. The current target recognizes these issues but does not make them auditable requirements.

D5. Distinguish human deception from automated-agent behavior. Disclosure may deter a person but have no such effect on a simple scanner or may be interpreted differently by an AI agent. Include separate actor classes, task budgets and software versions; do not aggregate them into one efficacy estimate.

D6. Measure inference provenance and incremental LLM contribution. The cited LLM paper aggregates per-IP histories and uses deterministic severity overrides. Its accuracy does not isolate model value from those rules; per-IP aggregation can merge separate users and separate one user across addresses. Compare rules-only, LLM-only and combined pipelines on held-out complete sessions, and test shared-address/rotating-identity benign fixtures. [R07]

D7. Public-value and adoption alternatives: operator interviews, mission downtime tolerance, training burden, existing control maturity and publication/IP expectations. Without a consenting mission owner, a technically

impressive range risks answering no adoption decision.

D8. Operating costs and end-of-life are missing from the target's economic table. Add patching/persona maintenance, independent validation, log retention, staff coverage, range refresh, receiving-party support and decommissioning. No credible business case can rest on capital/build cost alone.

D9. Evidence ledger remains a bibliography, not claim-level verification. Add exact title/authors/date/DOI, primary claim location, extraction depth, source type and contrary evidence for each material conclusion. Record query strings and inclusion/exclusion criteria; the report properly disclaims systematic coverage, so do not rebrand it as a systematic review.

E. Technical, adversarial and governance risks

E1. Evaluation invalidation by an unbeatable baseline. If all arms inherit a perfect deny policy and the only scored mission is prohibited routing, the ceiling effect hides any treatment difference. Also avoid a baseline so weak that decoys win only because conventional controls are omitted. Use a realistic fixed baseline, an outer safety boundary, and a mission deliberately testable inside the range.

E2. Confounded telemetry. Rotation can change log/session identifiers and break attribution, so lower observed attack rates can mean poorer observability. Include a ground-truth event stream and stable test IDs outside all treatment controllers. Test clock skew and log loss; an unsigned, missing or inconsistent record is unknown, never green.

E3. Policy-plane compromise and common-mode failure. Separate administrators are insufficient if authentication, hypervisor, signing keys, time or storage are shared. Map dependency concentration and inject benign faults below the controller. Safety authority must not depend on the same identity service that is under test. [Reviewer risk inference; consistent with R04's coordination findings.]

E4. Identity mediation bypass. Workload identities, automation/service accounts, CI/CD permissions, emergency overrides and federated delegation may reach genuine data without interactive network engagement. Test the application authorization path and the deployment pipeline, not only a decoy-facing gateway. High-confidence threat logic; no prevalence estimate is asserted.

E5. Cross-tenant/bystander leakage. Copying genuine records into a realistic decoy, placing secrets in logs, uncontrolled model context, cloud-backed analysis, and retained downloads can create a new exposure even if the attacker never escapes. Use synthetic data, destination-controlled capture and minimization. Pin model execution mode; "local LLM" alone is not a complete privacy assurance.

E6. Adaptive workload distraction. An actor can feed convincing but strategically irrelevant decoy behavior while a separate authorized-looking workflow affects the mission. Measure analyst queue length and simultaneous real-asset loss; engagement should not automatically elevate priority above stronger independent evidence.

E7. Automated response harm. A false classification may revoke a maintenance identity or interrupt a safety workflow. Require explicit per-action permissions and reversible automation; high-consequence actions need independent approval or a preapproved constrained policy. A human can also rubber-stamp erroneous output: measure decision quality and override effectiveness, not just existence of a button.

E8. Legal/owner boundaries. Counsel and each asset owner must approve scope, monitoring, retention and sharing; ONESTAMP is creative approval, not real-world legal authority. The related UIARC Project File expressly disclaims real-world privacy/surveillance authority and is not ratified. Its universal-admission framing must not be imported as a collection entitlement. [R13]

E9. Recovery and inter-organization accountability. Require a tested break-glass route that is available but audited, provider escalation contacts, restoration SLA assumptions, offline evidence/communications, joint abort authority and incident ownership. CISA's AAR documents failures of contacts, organizational role clarity, primary communications and cloud restoration understanding. A signed update alone cannot resolve those. [R04]

E10. Fundamental versus scoped feasibility. Ordinary decoys, protocol-specific proxies and managed reconfiguration are supported. General-purpose invisible transplantation of arbitrary third-party authenticated state is not established. Omniscient mapping/universal open-world closure is not a defensible measurable deliverable with unknown routes and actors. National ownership/authority cannot be engineered into existence by faster compute. These limits do not refute bounded stateful proxies or multisite federated research.

F. Proposed revisions by target section

Severity legend: critical = may invalidate the study or materially mislead investment; major = materially changes interpretation or readiness; minor = traceability/editorial issue. All below are proposed, not applied.

REV-C1-01 / CRITICAL / §§4,9: separate outer research containment from internal synthetic mission security; specify how the primary outcome can vary despite safety. Owner: research lead plus safety lead. Acceptance: diagram and adversarial ground-truth protocol distinguish both boundaries.

REV-C1-02 / CRITICAL / §10: regenerate narrative ranges from the base workbook; corrected numbers in Appendix 1. Owner: financial/model custodian. Acceptance: formula-linked outputs and independent unit checks match.

REV-C1-03 / MAJOR / §§2,9: preregister one primary mission endpoint, minimum worthwhile effect, censoring/competing-risk rules, clustering and multiple-comparison policy. Owner: methodologist. Acceptance: locked analysis plan before outcome collection.

REV-C1-04 / MAJOR / §9: replace generic 1-point benign-diversion tolerance with owner-approved absolute rate and harm criteria; add nondetection as failure/competing outcome rather than excluding it. Owner: mission owner/evaluator.

REV-C1-05 / MAJOR / §§3-5, Appendix A: replace categorical universal-migration prevention wording with scoped evidence status; add claim-source locations and retrieval-depth labels, including IEEE access failure. Owner: technical/source reviewer.

REV-C1-06 / MAJOR / §§5,7: add primary human disclosure and hybrid testbed evidence; branch learning postures rather than assume a sequence. Owner: research lead.

REV-C1-07 / MAJOR / §§9,14: stage human evaluation after ethics/consent/compensation and evaluator capacity, not automatically within 90 days. Owner: host institution and program lead.

REV-C1-08 / MAJOR / §10: add role-based resource model, recurring operating cost, dependency assumptions and whether contractor overhead already includes program overhead; do not double-count assessor time. Owner: financial lead.

REV-C1-09 / MAJOR / §11: explicitly restrict SEED to matching Dear Colleague Letter topics; label all other program pages as leads until a live matching solicitation is verified. Owner: grants office.

REV-C1-10 / MAJOR / §12: add common-mode dependency map, receiver-side independent acceptance, offline recovery and provider escalation rehearsals; treat governance evidence as a gate. Owner: safety/operations lead.

REV-C1-11 / MAJOR / §15: retain original canon verbatim in a separate baseline and present each reinterpretation as a numbered author decision, including any change from actual rekeying to display-only pulse. Owner: Nathan/document custodian.

REV-C1-12 / MINOR / §16, Appendix A: record exact search dates, queries, article versions and known publisher errata; distinguish framework evidence from outcomes. Owner: evidence custodian.

G. Experiment-by-experiment review and recommended validation

Common rule: all proposals below are isolated, synthetic, authorized tests. No external scanning, real credentials, uncontrolled malware or third-party network action. Publish sanitized fixtures and result schemas, not operational exploitation instructions. No experiment has been conducted by this review.

E1 Baseline increment — meaningful only after fixing the invariant. Separate baseline, baseline+static decoy, baseline+adaptive decoy and equal-budget conventional-upgrade arms. First an additive factorial test isolates component effect; then an equal-cost portfolio comparison asks the investment question. Those are different estimands and should not be confused. Predefine real-asset synthetic task success within a fixed window as primary; record all opportunities, failures, departures and safety aborts.

E2 Bypass population — stratify by new external session, valid user/workload authority, existing resident access, third-party application, management/update plane and nonnetwork maintenance tabletop. Sampling equal counts measures per-class capacity, not real incident-weighted risk. Report both unweighted class results and sensitivity to explicitly assumed threat weights. Do not extrapolate national incidence from a balanced range suite.

E3 Cadence — the target's factorial idea is sound but underspecified. Vary actuation layer separately from schedule. Match fixed and randomized policies on expected refresh count, log actual realized count, and record the detection information and runtime costs used by event-triggered policies. Event-triggered refresh is a response policy, so its timing cannot be naively treated as a randomized independent exposure. Include a sham UI pulse to isolate presentation effects from network changes. Prespecify the interaction tests or reduce the design to avoid underpowered many-cell comparisons.

E4 Provisioning — predeployment versus on-demand is useful. Match decoy content, warm/cold state and cache assumptions; start cost accounting before provisioning rather than when the service becomes ready. Record startup distribution, failed provisioning, rejected eligible sessions and idle reserve consumption, including no-attack intervals. Keep a fixed mission workload across arms.

E5 Learning — specify separate questions: attacker learning, defender updating, disclosure and model poisoning. Use an initial fixed-policy repeated-encounter study, then offline proposed updates assessed on a frozen holdout. Counterbalance participant experience/order, cluster by team/person, and prevent one participant seeing several supposedly blinded conditions without modeling carryover. Six postures are not six independent human samples. Poisoning belongs in an additional isolated candidate-policy test, not an automatic adaptation loop.

E6 Innocent traffic — include administrators, routine automation, shared-address users, partner monitors, atypical authorized tasks and a test emergency restoration. Measure absolute false diversion, severity-weighted harm, time to correction and lost work. Differentiate an extra authentication challenge from stopping a critical workflow. Zero events in a small sample does not establish a low population rate.

E7 Controller/update failure — add failures in shared identity, time, storage and management layers; missing/stale telemetry; controller disconnection mid-session; incompatible but correctly signed dummy package; attempted rollback to an insecure configuration; and receiver inability to reach the original supplier. Verifying a signature proves origin/integrity under key assumptions, not that content is safe. Test receiving-side behavior and independently preserved evidence.

Statistics revision: use intent-to-treat at scenario assignment, not only successful diversions. Time-to-isolation must include censored nondetections and competing mission loss/departure; use survival summaries or fixed-window outcomes rather than selected medians. Separate availability overhead from effective threat containment. Report uncertainty by attack class and human cluster; preregister a single primary contrast and adjust/explain secondary exploration. Sample sizes need pilot variance, minimum worthwhile effect and intracluster correlation. A pilot is not confirmatory evidence.

Additional recommended test G8 — disclosure and minimal-deception comparison. Compare decoys present/absent and disclosed/undisclosed in a controlled human cohort, informed by R10. Preserve genuine protections in all arms, include withheld mission scenarios, and score task progress plus analyst effects rather

than subjective intimidation alone. Ethics and consent must address disclosure/deception in the research itself.

Additional recommended test G9 — information provenance and ablation. Give a blinded evaluator rules-only, LLM-only and combined analytical outputs for held-out whole synthetic sessions, including prompt-injection-like untrusted text and benign noisy cases. Measure action correctness, calibration, coverage, analyst time and safety errors. This determines whether generative analysis is necessary at all. Do not assume zero-temperature generation guarantees reproducibility.

Numerical caution from R07: 0 false positives among 28 benign observations has an exact one-sided 95% upper binomial rate of about 10.15%, assuming independent representative trials. That is not a production bound, and clustering could further weaken interpretation. The reported zero is authentic but insufficient to justify low false-diversion budgets. This calculation critiques generalization, not the paper's explicit proof-of-concept framing.

H. Alternative architecture: resource-centric protection with minimal off-path deception

Alternative name: bounded resource protection plus isolated sentinel decoys. This is a recommended comparison design, not a deployed or validated OAL product.

1. Enforce resource-side identity, authorization, least privilege and segmentation for the genuine synthetic mission application; use existing deterministic controls rather than a generative national overlay.

2. Place a small set of static decoy resources/synthetic credentials only in approved paths. They have no production privileges, no live personal data and no authority to reconfigure protection.

3. Send decoy events and resource-side logs to an independent collector. Correlate using stable synthetic identities and disclose uncertainty. Human-reviewed response runs through a constrained operator playbook.

4. Refresh decoy content on operator-approved change or demonstrated staleness, not an assumed universal three-second rhythm. Keep the original visual pulse only in the creative/display track if Nathan approves that interpretation.

5. Release only versioned scoped configurations and evidence schemas, with receiving-owner acceptance, rollback and offline recovery. Optional LLM summaries are read-only and measured against deterministic summaries.

How it differs: no arbitrary established-session transplantation, no complete synthetic country, no continuously learning routing controller and no universal capture endpoint. It retains detection/intelligence opportunities while making real-resource protection independent of decoy realism. NIST's resource-centric model and documented decoy/tool functions support the component choices, not a measured outcome for this combination. [R01-R03]

Advantages (hypotheses): fewer cross-domain privileges, lower state/fidelity burden, clearer operator acceptance and easier causal attribution. Compromises: less immersive engagement, fewer toolchain observations, static-decoy detectability and continued need for identity/endpoint controls. Sentinel alerts still have selection bias and can be used as distraction. It may underperform adaptive engagement against some actors.

Comparison: three fixed-baseline arms (no deception, static sentinel, adaptive overlay) identify incremental component benefit; an equal-budget fourth arm reallocates adaptive-overhead savings to conventional hardening/response. Match mission workloads and threat opportunities, not only number of addresses. Compare synthetic mission loss, benign workflow harm, independent intelligence utility, analyst effort and lifecycle cost. If static/simpler arms meet the minimum useful benefit and adaptive does not add a worthwhile effect, retire adaptive modules without rewriting fictional canon.

I. Questions for the collaborative team

1. What is the primary estimand: lower synthetic mission loss, faster containment, better information, or lower workload? Which specific result would justify funding if the other outcomes do not improve?

2. How can unauthorized reachability vary between treatments while the independent deny invariant remains absolute? Draw both safety and mission-security boundaries and identify the ground-truth owner.

3. Which protocol and identity assumptions make a session eligible for redirection, and which established-state transfers are actually demonstrated by a cited implementation rather than inferred from proxy support?

4. Does anyone have publisher/full-text verification of IEEE article 11414228 and its 97-study denominator? Provide a stable DOI, exact title/authors and the locations supporting the evaluation-gap claims.

5. Which owner-approved benign workflows tolerate diversion, how is harm weighted, and what rate bound would be meaningful rather than merely statistically convenient?

6. Who owns and funds recurring operation, the independent verifier and receiving-party support if the original Architect/provider becomes unavailable?

7. Which real matching funding solicitation permits this applicant, research scope, budget and publication policy? Is SaTC SEED backed by an actual matching Dear Colleague Letter?

8. Under what data would the static/off-path alternative outperform adaptive immersion, and will the team accept that as a useful conclusion?

9. Which proposed interpretation changes locked creative behavior rather than clarifies implementation? Keep these as author decisions, not reviewer consensus.

J. Final recommendation and prioritized action plan

Priority 0 — preserve review independence. Archive the exact target and this review with hashes/version/date; gather the other five reviews before synthesis. Do not treat agreement among generated reports as replication. Do not revise canon or merge competing recommendations prematurely.

Priority 1 — correct deterministically verifiable errors. Repair §10 arithmetic and the associated executive/output ranges; add source-access status and SEED restrictions. Financial/evidence custodians should confirm corrected values independently. These corrections do not require running a cyber experiment.

Priority 2 — define the decision and estimand. Mission owner, safety lead and methodologist separate lab containment from challengeable synthetic assets; lock the primary outcome, benign harm budget, competing outcomes and minimum worthwhile effect. Document sessions that are ineligible for redirection.

Priority 3 — require institutional readiness. Establish consenting host, legal/privacy/ethics determinations, independent safety/ground-truth roles, resources and receiving-operator assumptions. No live traffic or production authority is implied by creative approval.

Priority 4 — run the smallest discriminating study only after authorization. Use static/no-deception/limited-adaptive arms in an existing isolated range. Begin machine replay and benign workflows; defer a substantial human repeated-game study until design and approvals justify it. Test E7 safety failure modes before adversarial workload. Obtain quotes or measured rates before escalating spending.

Priority 5 — decide by effect and precision. Proceed to a larger study only if a credible minimum beneficial effect remains plausible and harms are controlled, or there is a specific unresolved mechanism with a cheap discriminating test. Narrow to intelligence/analysis if that is the only supported benefit. Discontinue adaptive modules if a simpler equal-budget strategy dominates; do not call a small noisy null result definitive.

Proportionate spending: the target's base study budget is not demonstrably unreasonable, but four build stages should not be preauthorized. Scope a short asset-boundary/source/method design package first; price actual staff time and reuse a host's range when feasible. Tool license cost is not necessarily the dominant

expense; scientific evaluation, integration and ongoing support may dominate. No replacement monetary quote is supplied by this review.

Decision gates: no national deployment; no confirmatory experiment without revised protocol; no program-scale investment based on unverified bands; conditional support for bounded design/measurement. This changes the report's readiness assessment more than its overall cautious recommendation.

Independent answers to the three final questions

1. Strongest reason to proceed: the project can generate decision-quality evidence about when deception adds genuine mission protection and when nonengaging or trusted paths make apparent success misleading. A reusable, independently scored coverage method could prevent overinvestment and improve defensive choices even if NDIE is not built.

2. Strongest reason not to proceed: the current proposal may be a costly recombination of existing controls whose best outcomes are already delivered more simply, while perfect baseline isolation, selective engagement metrics and flexible endpoints make a convincing but scientifically uninformative demonstration likely. Without a concrete owner and causal question, a larger program is unjustified.

3. Single unresolved question most likely to change the recommendation: Under a realistic, specified mission and matched conventional controls, does adding adaptive deception produce a worthwhile reduction in mission loss or another explicitly chosen primary benefit, including when the attacker can avoid the decoys, at acceptable benign harm and lifetime cost? A precise positive result supports expansion; a precise negative result or simpler-strategy dominance supports narrowing or stopping.

Appendix 1. Independent budget reconciliation

The target uses base = (FTE x months/12 x loaded annual allowance + direct nonlabor) x 1.20. All four base totals reproduce. Its stated sensitivity low=0.75 x base and high=1.50 x base gives the following, rounded only for display:

Stage	Correct low (75%)	Base	Correct high (150%)
Desk	$98,325	$131,100	$196,650
PoC	$441,000	$588,000	$882,000
Range	$1,732,500	$2,310,000	$3,465,000
Multi-org	$6,093,000	$8,124,000	$12,186,000

Target narrative comparisons: PoC says about $397k-$794k, corrected $441k-$882k; range says $1.59m-$3.19m, corrected $1.7325m-$3.465m; multi-org says $5.44m-$10.89m, corrected $6.093m-$12.186m. Different unshown bases may have survived a previous revision, but that explanation is unverified. Fix via formulas and provenance, not by guessing intended budgets. These envelopes are sensitivity assumptions, not confidence or prediction intervals.

An additional point of false diversion on 10,000 benign opportunities corresponds to 100 additional diversions; exact consequence depends on task severity. For zero events in n independent Bernoulli trials, one-sided 95% upper rate = 1 - 0.05^(1/n); n=28 gives 0.1014657. Neither calculation is an NDIE measurement.

Appendix 2. Target citation verification status

Target S01 MITRE Deceive taxonomy: not separately refetched; companion S02 verified. Do not infer that every taxonomy entry or mapping was independently audited.

Target S02 Decoy Network Resource: verified at official technique page, R02. Definition and resource-exhaustion warning support limited cited role, not efficacy.

Target S03 NIST SP 800-160 v2r1: known from earlier research but not independently refetched in this cycle. No new certification of detailed page claims.

Target S04 NIST SP 800-207: official final publication/abstract rechecked, R01; no reliance on search's earlier draft.

Target S05 Cowrie: repository README verified, R03; experimental features/branches are not all production-assured. No pinned experiment commit provided by target.

Target S06 Chaudhry et al. 2026: publisher full-text sections retrieved, R06; systematic analyzed set 64, primarily low/medium scope. Citation valid, generalization needs scope labels.

Target S07 Cyber Storm IX AAR: official PDF text accessible in this review, R04; 2,200-plus and coordination claims verified. Prior screenshot restriction does not make parsed text unavailable now.

Target S08 Locked Shields: not independently refetched this cycle; do not upgrade organizer descriptions to comparative outcomes.

Target S09 Cyber Grand Challenge winners: not independently refetched this cycle; adjacent automation claim remains bounded, not fully audited here.

Target S10 AIxCC: official results and correction note verified, R05; 54/43/63 counts supported. Patch percentage wording on the page needs denominator care: 43/63 is about 68%, whereas 43/54 is about 80%. The target reports counts rather than the ambiguous percentage and is correct on that point.

Target S11 NSF SaTC: current official program verified, R08; RES/SEED ceilings and January/September target dates supported, SEED special-topic restriction must be explicit. Applicant-specific eligibility not determined.

Target S12 DARPA BAA index: not independently certified as a current matching award. A generic index is insufficient to verify availability or eligibility.

Target S13 DOE SBIR: not independently certified for active matching call, topic or applicant in this review.

Target S14 NIST SBIR: not independently certified for active matching call, topic or applicant in this review.

Target S15 IEEE 11414228: publication existence/title corroborated via indexed publisher result; direct publisher fetch failed at abstract/document variants. A secondary abstract mirror corroborates 97 included studies and performance gaps; full paper and study quality were not verified. R09 marks partial status, not fabrication.

Target S16 Oliveira et al. 2026: publisher text verified, R07; single-run/52 observations supported. Cautions about observation independence, deterministic overrides and per-IP aggregation remain.

Target S17 NIST AI 100-2e2025: official publication page verified, R13a. March 2025 taxonomy and poisoning/evasion scope supported; page notes an identified error and corrected PDF history. No specific mitigation validated by citing taxonomy alone.

Target S18 BLS: not refetched in this cycle; quoted wage is not the operative budget input and does not validate loaded contractor costs. No new independent verification asserted.

No verified citation in this audit was shown fabricated. One technical reference remains only partly verified, several were not rechecked, and matching funding calls remain uncertified. Exact verification depth should remain visible rather than compressing all statuses into "sources checked."

Appendix 3. Review source ledger

All online sources below accessed November 5, 2026; publication dates are stated where verified. Target section references refer to the supplied ten-page PDF.

R01 — NIST SP 800-207, Zero Trust Architecture, August 2020. Official final publication abstract. https://csrc.nist.gov/pubs/sp/800/207/final . Supports resource-centric/authentication scope; not an efficacy trial.

R02 — MITRE D3FEND, Decoy Network Resource D3-DNR, undated current page. https://d3fend.mitre.org/technique/d3f%3ADecoyNetworkResource/ . Definition, considerations, experimental mapping and patent/functionality caution.

R03 — Cowrie project README, current unpinned repository view. https://github.com/cowrie/cowrie . Emulated shell, proxy, experimental LLM modes; developer documentation not independent outcome study.

R04 — CISA Cyber Storm IX After-Action Report, September 2024. https://www.cisa.gov/sites/default/files/2024-10/Cyber%20Storm%20IX%20After-Action%20Report%20v00%2020241001_508.pdf . Executive summary pp.1-2; overview pp.3-4; cloud/internal-process findings pp.5-10. Exercise/feedback evidence, not randomized evaluation.

R05 — DARPA, AI Cyber Challenge marks pivotal inflection point for cyber defense, August 8, 2025. https://www.darpa.mil/news/2025/aixcc-results . Final competition findings and editor's corrected denominator; organizer account, not an NDIE trial.

R06 — Chaudhry, Andersen and Dragoni, A Review of Honeypots: Fingerprinting Techniques, Detection, and Evasion Mechanisms, Future Internet 18(4),190, April 1,2026. DOI 10.3390/fi18040190. https://www.mdpi.com/1999-5903/18/4/190 . Methods §§3.3-3.5 and scope/summary; secondary synthesis with bounded inclusion.

R07 — Oliveira, Gomes and Almeida, An Integrated Honeypot and LLM-Based Framework for near Real-Time Detection and Behavioral Analysis of Malicious Activities, Journal of Cybersecurity and Privacy 6(5),147, August 31,2026. DOI 10.3390/jcp6050147. https://www.mdpi.com/2624-800X/6/5/147 . Abstract and §§3.1-3.4; controlled proof of concept, strong deterministic rules, not production specificity estimate.

R08 — NSF Security, Privacy, and Trust in Cyberspace program, guidelines NSF 25-515; program page last updated October 5,2026.
https://www.nsf.gov/funding/opportunities/nsf-satc-us-national-science-foundation-security-privacy-trust . RES/SEED scope and targets; not applicant approval.

R09 — Moving Target Defense in Software-Defined Networks — A Systematic Review, IEEE Access 2026, indexed DOI 10.1109/ACCESS.2026.3668537; indexed author listing Larissa Souto et al. https://ieeexplore.ieee.org/document/11414228 . Publisher fetch failed. Secondary mirror https://discovery.researcher.life/topic/moving-target-defense/967934?page=1 reproduces abstract with 302 records/97 studies. Title/existence verified, full text and exact counts only partly corroborated; no invented author list supplied.

R10 — Ferguson-Walter, Major, Johnson and Muhleman, Examining the Efficacy of Decoy-based and Psychological Cyber Deception, USENIX Security 2021, pp.1127-1144.
https://www.usenix.org/conference/usenixsecurity21/presentation/ferguson-walter . Publication/abstract verified; over 130 red teamers/four conditions/disclosure result. Full effect reanalysis not performed.

R11 — Kim, Cho, Lim, Moore, Nelson, Ko and Kim, Evaluating Performance and Security of a Hybrid Moving Target Defense in SDN Environments, IEEE QRS 2022. DOI 10.1109/QRS57517.2022.00037. Institutional metadata/abstract: https://sos-vo.org/publications/evaluating-performance-and-security-hybrid-moving-target-defense-sdn-environments . Physical testbed and hybrid triggering claimed by authors; detailed effect sizes not verified.

R12 — Casola, De Benedictis and Iorio, Moving target defense in Edge-Cloud systems: a systematic literature review, Journal of Cloud Computing, June 15,2026.
https://link.springer.com/article/10.1186/s13677-026-00920-7 . Publisher search extraction only; overhead discussion, not substitute for R09 or new full-text verification.

R13a — NIST AI 100-2e2025, Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations, March 2025. https://csrc.nist.gov/pubs/ai/100/2/e2025/final . Official abstract and errata/planning notes, not deployment guarantee.

R13b — UIARC_Preliminary_Information_Collection_Charter_Perplexity.pdf, August 23,2026, Project File. Page 1 and Parts III/VII/IX excerpts: preliminary, fictional, not ratified, no real-world authority. No external legal policy inferred.

Appendix 4. Audit trail and limitations

Files read for the first-pass review: target research PDF only; previous source-dossier/prompt context retained. Other attached research executions not read. Related project-file excerpt consulted solely to prevent fictional authority spillover.

New verification operations: batched fetches of the target's 2026 honeypot review, LLM paper, IEEE page, AIxCC results, NSF program, Cyber Storm AAR; then NIST final zero-trust publication, Cowrie and USENIX human-study page; IEEE retried once at alternate document URL plus D3FEND and NIST AML. IEEE access failed both times; indexed/search excerpts were labeled as such.

New queries: "Examining Efficacy Decoy Psychological Cyber Deception 130 red teamers Ferguson Walter"; "NIST zero trust architecture policy enforcement point resource access SP 800-207"; "11414228 97 moving target"; "Moving Target Defense SDN 2026 systematic review". Unrelated search results were excluded; the later 2026 Edge-Cloud survey and 2022 hybrid testbed were retained as bounded leads. This is targeted verification, not exhaustive newer-literature coverage.

Python performed local SHA-256/10-page check and independent budget/zero-event interval calculations. No simulated attack, network measurement, trial enrollment, laboratory run, performance re-estimation from raw papers or hypothesis test was conducted. All proposed experimental changes remain unexecuted and unapproved.

Version history: v1.0 is this first Cycle 1 independent critical review. Corrections to this review should be versioned separately, retaining original target hash and review provenance. Next round should record reviewer, disputed claim, exact proposed wording, source location, severity, whether it affects the verdict, and owner disposition. No source canon, research report, architecture or repository was modified by preparing this document.